Availability: One important failure statistic **Availability** the percentage of time the networks available to users

Charge-back policy: to allocate the costs associated with the network to specific users.

Configuration management: means managing the network's hardware and software configuration, and documenting it, and ensuring it is updated as the configuration changes

Cost management: One of the most challenging areas of network management over the past few years has been

Desktop management: sometimes called electronic software delivery or automated software delivery, is one solution to the configuration problem

Downtime: includes times when the network is unavailable because of faults and routine maintenance and network upgrades

End user support: means solving whatever problems users encounter while using the network.

Fault management: means preventing, detecting, and correcting faults in the network circuits, hardware, and software

Firefighting: Unfortunately, many network managers spend most of their time **firefighting** dealing with breakdowns and immediate problems

Help desk: the organization should maintain a central telephone number for network users to call when any problem occurs in the network

Logical network Parameters: include performance measurement systems that keep track of user response times, the
Volume of traffic on a specific circuit,

Mean time between Failures (MTBF): is the number of hours or days of continuous operation before a component fails.

Mean time to diagnose (MTTD): is the average number of minutes until the root cause of the failure is correctly diagnosed

Mean time to fix (MTTF): Finally, after the vendor or internal support group arrives on the premises, the last statistic

Mean time to repair (MTTR): is the average number of minutes or hours until the failed device or circuit is operational again

Mean time to respond (MTTR): is the average number of minutes or hours until service personnel arrive at the problem location to begin work on the problem

Monitor: Most large organizations and many smaller ones use network management software to **monitor** and control their networks

Network cost of ownership (NCO): Such measures, often called **(NCO)** or real TCO, have found that NCO ranges between $1,500 and $3,500 per computer per year

Network documentation: Configuration documentation includes information about network hardware, network software, user and application profiles, and network documentation

Network management: is the process of operating, monitoring, and controlling the network to ensure it works as intended and provides value to its users

Network monitoring: Both require **network monitoring,**
Which means keeping track of the operation of network circuits and devices to ensure they are functioning properly and to determine how heavily they are used.

Network operations center (NOC): that is responsible for monitoring and fixing problems

Network weather map: One common type of monitoring approach is the Network weather map which displays the usage of all major circuits in the network in real time

Performance management: means ensuring the network is operating as efficiently as possible whereas

Physical network Parameters: includes monitoring the operation of the network's modems, multiplexers, circuits linking the various hardware devices, and any other network devices

Problem statistics: are important because they are a control device for the network managers as well as for vendors

Quality control chart: Such predictions can be accomplished by establishing simple **quality control charts** similar to those used in manufacturing

Service-level agreement (SLA): their common carriers and Internet service providers

Total cost of ownership (TCO): is a measure of how much it costs per year to keep one computer operating
Trouble ticket: The reports they produce are known
Uptime:

Problem tracking: allows the network manager to determine who is responsible for correcting any outstanding problems
_ Desktop management sometimes called **electronic software Delivery** or automated
_ **Network software:** Configuration documentation includes Information about network hardware,
_ **Problem prioritizing:** helps ensure that critical problems get Priority over less important ones

Q 2) compare between physical network and logical network

1) **physical network parameters**

A_ includes monitoring the operation of the networks Modems
B_ multiplexers
C_ circuits linking the various hardware devices

2) **Logical network parameters**

A_ keep track of user response times
B_ the volume of traffic on a specific circuit

Q 3) help disk (it have not been decentralized into user Department: explain why;
Is a proliferation of help desks and support staff members, many of whom tend to be generalists rather than specialists in one area?

Q 4) اختصارات

ESD: electronic software delivery

DMI: Desktop Management Interface
NOC: network operations center
MTBF: mean time between failures
PVC: permanent virtual circuit
SLA: Service-level agreement
TCO: Total cost of ownership
NCO: Network cost of ownership
MTTR: Mean time to respond
MTBF: Mean time between Failures
MTTD: Mean time to diagnose
MTTF: Mean time to fix

Q 5) الموازنة اعلى قيمة واقل قيمة

Electronic software delivery
High: WAN Operations 7,410,000
Low: Policy Management 75,000

End User Support
High: Support Center (Help Desk) 2,741,000
Low: Student Residence Halls Support 279,000

Client Hardware
High: Student Technology Centers Equipment and Supplies 2,456,000
Low: Student Residence Halls Equipment and Supplies 601,000

Application Software
High: Software Site Licenses 2,540,000
Low: Student Residence Halls Software 146,000